

"We are treated as if we don't deserve to be here"

First-year graduate students' accounts of their struggles in a doctoral program

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INTRODUCTION & LITERATURE REVIEW

- First-year students in a physics doctoral program may face many challenges, including:
 - Mathematically rigorous coursework [1]
 - Serving as teaching assistants (TAs)--often for the first time and without adequate training [2]
 - Finding a research advisor [3, 4]
 - Taking a written qualifying examination (QE) [5, 6]
- Recent survey of 255 physics & astronomy (P&A) graduate students found [7]:
 - 38.8% reported moderate to severe anxiety
 - 35.3% reported moderate to severe depression
 - One in five reported recent suicidal ideation
 - With loneliness, advisor support, and gender found as significant predictors
 - Advisor engagement, social connections, and sense of professional progression are important to foster graduate student wellbeing
- Advisors should normalize struggle & failure, validate students' abilities, and discuss issues of race and gender in academia [8]
 - Academic difficulty does not equal evidence of inadequacy
- Previous study at Auburn University found graduate students believed the QE served as a gate-keeping mechanism [9]

METHODS

- 2023: 9 of the 11 students
- 2026: 13 of the 22 students
- General thematic analysis [10]

2023 Cohort Challenges (N = 9)	
Main Challenges (n)	Sub-challenges (n)
Faculty interactions (5)	Lack of support (4) Lack of communication (3) QE value statements (3) Lack of respect (3) Valuing research over teaching (2) Lack of educational standards (2) Course instructor value statements (1)
Structural (5)	Coursework & course structure (5) QE structure (3) GPA requirement (1) Balancing responsibilities (1)
Financial (2)	Health insurance & student fees (2) Relocation/moving costs (1) Low income (1)
Emotional (2)	Loneliness (2) Lack of breaks/work-life balance (1) Disregard for well-being/burn out (1)
Personal (1)	Family (1)

INSTITUTIONAL CONTEXT

- Auburn University Department of Physics Doctoral Program 1st Year:
 - Four core courses: Classical Mechanics, Electricity & Magnetism, Quantum Mechanics, Statistical Mechanics
 - 2 elective courses
 - 1 credit pedagogy course, 1 credit research seminar course, attendance at all colloquia
 - Work as TAs
 - GPA ≥ 3.0 (probation and potential suspension for students below this threshold)
 - Must take written QE at end of first summer
 - One exam for each core course, 210 minutes each
 - Minimum passing score ~50% (some downward, but arbitrary, flexibility in the threshold)
 - Two attempts on each exam allowed
 - Unsuccessful students are dismissed from the PhD program with the option to receive a terminal master's degree

"FIRST, WE'D LIKE TO GIVE YOU THE OPPORTUNITY TO BRIEFLY DESCRIBE THE CHALLENGES YOU HAVE FACED AS A FIRST-YEAR GRADUATE STUDENT, WHETHER OR NOT THEY ARE RELATED TO THE [QE]."

"I didn't really feel supported by faculty, with the exception of a few specific people."

"[It] can be disheartening when important information doesn't reach us in a timely manner, causing unnecessary stress and hindering our ability to navigate through obstacles."

"The [QE] serves more as a filter, endurance test & hazing tradition rather than a fair assessment."

"We even had a professor post-exam say... 'maybe you shouldn't be in graduate school.'"

"[The QE] primarily gauges theoretical knowledge, which is unfair and unreasonable for students pursuing experimental or other research... prioritizing stamina and endurance rather than serving as a genuine assessment."

"Students are compelled to spend their entire summer isolated, studying relentlessly."

2026 Cohort Challenges (N = 13)	
Main Challenges (n)	Sub-challenges (n)
Faculty interactions (5)	Lack of educational standards (3) Lack of respect (2) Lack of communication (1)
Structural (6)	Coursework & course structure (4) QE Structure (2) Balancing responsibilities (1) Teaching (1)
Emotional (3)	Loneliness (2) Lack of breaks/work-life balance (1) Disregard for well-being/burn out (1)
Peer interactions (3)	Exclusionary behaviors (3) Cohort cohesion (1)
International student challenges (2)	Weather (1) Relocation to U.S. (1) Insufficient UG coursework (1)
Transition to grad school (1)	Different expectations (1) Different teaching style (1)
Personal (1)	Family (1)

"Having a bad instructor has such a cascading effect on learning experience," "one professor did not seem to value [their] time teaching graduate students, which made the first [year] significantly less welcoming."

"Vague information and veiled threats about our future without concrete information create an atmosphere of fear and a sense of having a lack of permanence that makes establishing oneself in the department or university difficult, especially when student confidentiality is not maintained by faculty."

"Lack of enforcement of a code of conduct has caused classmates to behave in exclusionary and prejudiced manners that isolate certain students without a solid path of recourse."

"I am commonly at [department building] from 8am to 10pm every day just to get all of my work done in a timely manner...So, I guess the main difficulty is just the mental battle of persisting on and not getting burntout by the work and the [QE] always looming over you."

DISCUSSION & CONCLUSION

- Faculty and structural challenges present for both cohorts → Peers are a new challenge in 2026
 - 2026 cohort is twice the size but nature of peer-related challenges are concerning
 - Future work should examine conditions that shape peer dynamics in graduate cohorts
- Consistent concerns on coursework and instructional practices
 - Future work should examine pedagogies used in graduate-level courses and barriers to adopting research-based practices
- Loneliness was present for both cohorts, particularly concerning with findings that loneliness is a significant predictor of graduate student well-being [7]
- GPA threshold and restrictions on number of QE attempts create high stakes environment
 - Constrains opportunities to normalize failure & reinforces perceptions of failure
 - Reducing structural pressure may be more effective than individual level anxiety interventions [11]
- 2023 student-driven efforts are a first order change
 - Second order change would require altering assessment and professional development system in the program
- We are using the "Four frames" model of change to understand the people, power, symbols, and structures which need to be targeted to create more sustainable change [12]

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